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Importance of the type of soil for the induction of micronuclei and the growth of primary roots of *Vicia faba* treated with the herbicides atrazine, glyphosate and maleic hydrazide

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Summary

Research was carried out on the genotoxic effects (induction of micronucleated cells in primary root tips) and toxic effects (reduction in primary root growth) in young plants of *Vicia faba* grown in soils with different organic matter contents and treated with the herbicides atrazine, glyphosate and maleic hydrazide. The data obtained show that the genotoxic effects are noticeably influenced by the interactions between the herbicide and the type of soil in which the *Vicia faba* have grown. While maleic hydrazide proved to be highly clastogenic for young plants grown in both soils, atrazine was genotoxic only in young plants grown in soil poor in organic matter. Glyphosate did not induce micronuclei under either soil condition, but induced a significant toxic effect.

Several factors affect the distribution and the fate of herbicides in the soil. An important interaction is the capability of the soil to adsorb the herbicide. This capability varies with different soils.

The purpose of this paper was to study the genotoxic effects (induction of micronuclei) in

root tips of *Vicia faba*, grown in different types of soil and treated with maleic hydrazide, glyphosate or atrazine.

Atrazine is an *s*-triazine which selectively inhibits chlorophyll photosynthesis at the Hill reaction level. It is known from the literature that plant activation is a requirement for the metabolic conversion of atrazine to a mutagen (Loprieno and Adler, 1980; Adler, 1980; Plewa et al., 1984; Means et al., 1988).

Glyphosate is a non-selective phosphoric organic chemical, which acts primarily as a competi-

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tive inhibitor of 5-enolpyruvylshikimic acid 3-phosphate synthase (Amrhein et al., 1980), an enzyme essential to the synthesis of aromatic amino acids in plants. The potential genotoxicity of glyphosate is questionable. Negative results have been observed in *Salmonella*, in *E. coli* and in *Drosophila* (Chiao et al., 1980; Shirasu et al., 1982; Gopalan and Njagi, 1981; Li and Long, 1988). However, glyphosate at high concentrations caused a significant increase in SCE in human lymphocytes in vitro with no dose-response relationship (Vig Fusson and Vyse, 1980).

Maleic hydrazide is a pyridazine which inhibits the synthesis of nucleic acids and proteins. It induces chromosome aberrations and SCE in plants (Kihlman, 1956; Kihlman and Sturelid, 1978). Results on clastogen effects are contradictory in animals, their sensitivity to maleic hydrazide being much lower compared to that found in plants (Perry and Evans, 1975).

Materials and methods

Seeds of *Vicia faba* were removed from cold storage, and germinated in aluminum basins containing two different types of soil, an agricultural one of low organic matter content (typic udifluvents, named Piedifiume) and a forest soil of high organic matter content (typic melanudal, named Vico). Table 1 shows the main chemical and physical characteristics of the two soils.

200 ml of water, in which the different herbicides had been dissolved, was added to each basin containing 600 g of soil. Atrazine was used as an emulsifiable liquid in Fogard trading formulation (SIAPA) containing 45% of active agent. The active doses of atrazine used were: 175, 350,

700, 1750, 3500, 5250 and 7000 $\mu\text{g/g}$ of soil. Glyphosate was used as an emulsifiable liquid in Solado trading formulation (SIAPA) containing 21% of active agent. The active doses used were: 35, 70, 105, 140, 350, 700, 1050 and 1400 $\mu\text{g/g}$ of soil. Maleic hydrazide was used as pure product (United States Biochemical Corporation) and dissolved in water at concentrations of 2.8, 5.6, 8.4 and 11.2 $\mu\text{g/g}$ of soil.

The different samples were kept in a climatic chamber at $20 \pm 1^\circ\text{C}$. After 4 days of treatment the young plants were removed, the primary root length was measured and the root tips were fixed in ethyl alcohol and acetic acid (3:1, v/v). Before being squashed under a cover slip, the samples were treated with pectinase (1%) at 38°C for 30 min. Permanent slides were mounted in Canada balsam. Duplicate experiments were carried out for each treatment.

A total of 20,000 cells were analyzed, isolated from 10 different root tips (2000 cells per root tip) for each herbicide concentration. The micronuclei were counted only when their diameter did not exceed 1/3 of the main nucleus and when they were localized inside the cell wall and in the cytoplasmic area surrounding the main nucleus.

The data reported (means and standard deviations) were analyzed using a one-way analysis of variance procedure and compared at the $p = 0.05$ level.

Results

Tables 2, 3 and 4 present the data for the doses of the atrazine, glyphosate and maleic hydrazide.

The data reveal different responses according to the herbicide examined. Atrazine shows a direct correlation between dose and micronucleated cells only in Piedifiume soil. Glyphosate did not increase the number of micronucleated cells. Maleic hydrazide shows a good correlation between dose and induction of micronucleated cells in both soils, although this induction is greater in Piedifiume soil.

Tables 2, 3 and 4 also show growth values of the primary roots after 4 days of treatment with each herbicide and soil type. While no significant reduction of primary root growth is observed with

TABLE 1
CHARACTERISTICS OF THE SOILS USED

	Piedifiume	Vico
Sand (%)	29.1	45.3
Silt (%)	55.3	39.3
Clay (%)	15.6	15.4
pH	8.1	5.1
Organic matter (%)	2.4	14.5
C.E.C. (meq/100 g) ^a	17.88	50.20

^a C.E.C., cation exchange capacity.

TABLE 2

FREQUENCY (%) OF MICRONUCLEATED CELLS (MC) AND VALUES (mm) OF PRIMARY ROOT GROWTH (L) OF *Vicia faba* AFTER 4 DAYS OF TREATMENT WITH ATRAZINE

Treatment (ppm)	Piedifume		Vico	
	MC	L	MC	L
Control	0.26 ± 0.29	28.9 ± 5.26	0.14 ± 0.23	36.4 ± 7.23
175	0.27 ± 0.40	30.7 ± 6.39	0.14 ± 0.22	37.2 ± 6.16
350	0.57 ± 0.48	31.0 ± 6.84	0.15 ± 0.24	36.0 ± 7.79
525	0.66 ± 0.34	33.7 ± 6.57	0.14 ± 0.22	36.2 ± 7.98
700	1.45 ± 0.91	28.4 ± 4.40	0.14 ± 0.23	38.2 ± 8.39
1750	2.19 ± 0.74	27.8 ± 4.69	0.18 ± 0.23	39.4 ± 7.18
3500	3.27 ± 1.71	28.2 ± 5.31	0.18 ± 0.24	38.6 ± 7.76
5250	3.67 ± 1.88	27.8 ± 6.11	0.12 ± 0.19	38.3 ± 6.18
7000	4.28 ± 1.19	25.1 ± 3.31	0.14 ± 0.24	34.8 ± 6.09
Analysis of variance	$F = 22.25$ $p < 0.01$	$F = 1.95$ NS	$F = 0.75$ NS	$F = 0.42$ NS

The frequency of MC was calculated on 20,000 root-tip cells (10 root tips, 2000 cells per root tip) for each herbicide concentration.

atrazine even at much higher doses than those given in the field (up to 700 ppm vs. 1–2 ppm), there was a significant decrease in primary root growth values when glyphosate was used. For maleic hydrazide a reduction in primary root growth was observed only for Piedifume soil. In all the samples the young plants grown in Vico

soil always showed primary root growth values higher than those in Piedifume soil.

Discussion

We have previously observed (De Marco et al., 1990) that the herbicide alachlor does not induce

TABLE 3

FREQUENCY (%) OF MICRONUCLEATED CELLS (MC) AND VALUES (mm) OF PRIMARY ROOT GROWTH (L) OF *Vicia faba* AFTER 4 DAYS OF TREATMENT WITH GLYPHOSATE

Treatment (ppm)	Piedifume		Vico	
	MC	L	MC	L
Control	0.32 ± 0.44	37.8 ± 9.03	0.29 ± 0.34	43.6 ± 9.17
35	0.36 ± 0.67	32.1 ± 7.75	0.33 ± 0.37	40.2 ± 9.69
70	0.69 ± 0.90	24.9 ± 5.49	0.25 ± 0.45	38.5 ± 11.35
105	0.88 ± 0.58	23.3 ± 4.59	0.47 ± 0.72	38.3 ± 12.64
140	0.77 ± 0.45	22.3 ± 3.94	0.76 ± 0.80	38.0 ± 9.54
350	0.91 ± 0.70	21.4 ± 3.31	0.97 ± 0.64	37.5 ± 8.37
700	1.07 ± 0.72	21.1 ± 5.63	0.97 ± 0.83	31.3 ± 5.52
1050	1.14 ± 0.87	19.2 ± 1.13	1.01 ± 1.05	30.5 ± 4.99
1400	1.11 ± 0.91	18.9 ± 2.64	1.26 ± 1.03	26.6 ± 6.34
Analysis of variance	$F = 1.62$ NS	$F = 13.96$ $p < 0.01$	$F = 1.81$ NS	$F = 2.51$ $p < 0.01$

The frequency of MC was calculated on 20,000 root-tip cells (10 root tips, 2000 cells per root tip) for each herbicide concentration.

TABLE 4

FREQUENCY (%) OF MICRONUCLEATED CELLS (MC) AND VALUES (mm) OF PRIMARY ROOT GROWTH (L) OF *Vicia faba* AFTER 4 DAYS OF TREATMENT WITH MALEIC HYDRAZIDE

Treatment (ppm)	Piedifume		Vico	
	MC	L	MC	L
Control	0.46 ± 0.40	31.6 ± 7.24	0.33 ± 0.41	35.1 ± 7.82
2.8	45.64 ± 12.56	27.7 ± 3.71	5.12 ± 3.38	33.8 ± 5.92
5.6	61.77 ± 25.53	25.2 ± 3.46	12.20 ± 3.83	30.2 ± 6.00
8.4	74.23 ± 28.60	24.1 ± 4.48	26.19 ± 4.66	32.0 ± 3.29
11.2	109.19 ± 46.68	23.9 ± 3.84	33.37 ± 6.09	34.0 ± 4.42
Analysis of variance	$F = 20.18$ $p < 0.01$	$F = 4.60$ $p < 0.01$	$F = 24.30$ $p < 0.01$	$F = 0.95$ NS

The frequency of MC was calculated on 20,000 root-tip cells (10 root tips, 2000 cells per root tip) for each herbicide concentration.

significant increases of micronucleated cells in soil with high organic matter content, while a dose- and time-dependent increase was observed in soil with low organic content. HPLC analysis carried out on the concentration of alachlor present in both kinds of soil at the end of treatment showed different degrees of adsorption of the two soils. The soil rich in organic matter had a higher capability of adsorption for alachlor.

In this work we did not observe any induction of micronucleated cells by atrazine in root tips of the young plants grown in soil rich in organic matter (Vico), while in soil poor in organic matter (Piedifume) there was induction of micronucleated cells by atrazine which was dependent on the dose of the treatment. In the literature a close correlation between adsorption of the herbicide and organic colloid content has been demonstrated (Stevenson, 1982). From these results we may deduce that where the adsorption by the soil is high (Vico) the presence of atrazine in the circulating solution is low and therefore a very small amount will transfer into the plant.

Also maleic hydrazide is strongly adsorbed by organic colloids; however, because of its high clastogenic capacity it induced micronucleated cells in root tips of plants grown in both soils, but less in organic soil (Vico), although Vico soil has a more acid pH (5.1) and maleic hydrazide is more genotoxic in *Vicia faba* and *Allium cepa* at lower pH (Kihlman, 1956; Cortes et al., 1987).

With glyphosate no significant differences in induction of micronucleated cells were found between the two soils. These observations agree with the adsorption of glyphosate as reported in the literature. This adsorption is completely independent from the organic matter content of the soil (Hance, 1976), so that similar amounts of herbicide should be present in the circulating solutions of both soils; thus glyphosate was not genotoxic in this study.

In experimental conditions like ours, tests on micronuclei and primary root growth give results for genotoxicity and toxicity that cannot always be correlated. Maleic hydrazide was shown to be a strong mutagen for *Vicia faba*, being active at very low doses (2.8 ppm) where there is a decrease in root growth only in Piedifume soil. Atrazine was a weak clastogen and induced a significant increase in micronucleated cells only at very high doses (from 700 ppm). However, it is very persistent in the environment and therefore may accumulate and be bioconcentrated. Primary root growth was not reduced even at the highest doses (up to 7000 ppm).

Glyphosate was not clastogenic and induced a significant reduction in root growth (from 350 ppm) at doses higher than those used in the field (1–2 ppm). Finally, we suggest that the experiment presented here can be used as a model for the study of the environmental risk connected with the increasing use of herbicides in agricul-

ture; this model, in fact, is easy to carry out and provides information on two processes: toxicity and genotoxicity.

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